Reserve Bank of India & Banking Technology Report:
The Impact of Generative Artificial Intelligence on the Indian Banking Sector (2024-2029)

1. Executive Overview
The Indian commercial banking sector, comprising scheduled commercial banks (SCBs) including State Bank of India, HDFC Bank, ICICI Bank, and Axis Bank, is undergoing rapid digital transformation. Generative AI is anticipated to contribute between billion to billion in aggregate operating value to Indian financial services over the next five years.

2. Strategic Core Banking and Front-Office Opportunities
- Vernacular Conversational Banking: Delivering natural language financial advisory across 12+ official Indian languages (Hindi, Tamil, Telugu, Marathi, etc.) to drive financial inclusion in tier-2 and tier-3 cities.
- Automated Credit Underwriting and Documentation: Reducing MSME loan sanction turnaround times (TAT) from 4 business days to under 45 minutes using automated financial statement parsing and risk synthesis.
- Personalized Wealth Advisory: Algorithmic portfolio recommendations and proactive tax saving insights tailored to retail depositors.

3. Regulatory Compliance and RBI Governance Directives

The Reserve Bank of India (RBI) has articulated strict supervisory guidelines regarding AI and Machine Learning deployment:

- Model Explainability: Banks cannot deploy 'black-box' deep learning models for algorithmic lending or credit rejection without verifiable explainability (XAI) audits.
- Data Sovereignty and Local Storage: Financial transaction logs and Indian citizen Personally Identifiable Information (PII) must strictly reside within domestic data centers located inside the borders of India.
- Human-in-the-Loop Safeguards: Critical customer-facing credit decisions, account freezing, and suspicious transaction reporting (STR) require mandatory human sign-off.

4. Systemic Risks, Deepfakes, and Operational Challenges

- Synthetic Fraud & Deepfake Impersonation: Adversarial AI is being leveraged to spoof video KYC (v-KYC) processes and voice authentication systems, necessitating liveness detection upgrades.
- Core Banking Legacy Debt: Legacy core-banking backends (Finacle, BaNCS) require substantial API middleware modernization before enterprise LLMs can securely access real-time ledgers.
- Talent Shortage & Training Costs: The acute shortage of specialized AI safety engineers and data compliance officers in public sector banks remains an operational bottleneck.